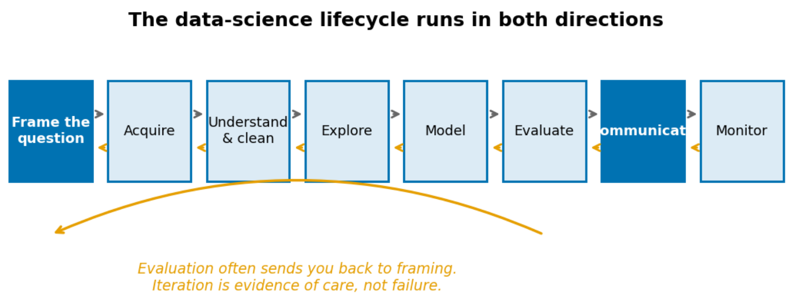
NEW UZBEKISTAN UNIVERSITY

Week 1
Thinking Like a Data Scientist

A useful analysis begins with a question, not with an algorithm.



Chapters 1–2 • Lecture

Today’s habit: ask what one row means before calculating anything.

TODAY

Two hours: from a vague concern to a defensible question

0–15	Welcome, goals, and course habits
15–45	Questions, claim types, and the data-science lifecycle
45–60	Rows, variables, and a short check
60–70	Break
70–110	Sources, samples, proxies, and joins
110–120	Synthesis, exit ticket, and Lab 1 handoff

OPENING QUESTION

When have you seen a number used to make a decision? Did the number deserve your trust?

What this course expects from you

CLO1

Explain

data collection, preprocessing, EDA, and basic statistics

CLO2

Prepare

numerical, categorical, time-series, and text data

CLO3–4

Apply & communicate

basic ML plus responsible visual, dashboard, and report design

CLASS NORM

A result is not complete until you explain what it means and name at least one limitation.

COURSE PRACTICE

How your learning will be assessed

Assessment	When	Weight
Midterm exam	Week 8	**20%**
Final exam	Week 16	**20%**
Weekly labs & in-class tasks	Weeks 1–14	**15%**
Data Quality Report	Week 4	**10%**
EDA & Visualization Brief	Week 7	**10%**
Modelling Assignment	Week 13	**10%**
Capstone + dashboard + presentation	Week 15	**15%**

QUICK REFLECTION

Which assessment will test whether you can do data science—not merely repeat definitions?

AI, ML, deep learning, and data science

Related, but not identical.

Artificial Intelligence

Broad goal: build systems that perform tasks associated with intelligent behaviour.

Machine Learning

A part of AI: learn patterns from data to make predictions or decisions.

Deep learning = neural-network ML

Data Science

Answers questions with data, quantifies uncertainty, and supports decisions.

It may use ML or deep learning. It also includes cleaning, statistics, visualisation, experiments, and dashboards.

CHECK YOUR UNDERSTANDING

Which is data science without machine learning: predicting withdrawal, or comparing satisfaction across programmes with a chart?

START WITH A PROBLEM

A concern is not yet an analytical question

The Dean says:

“Students seem disengaged this year. Can you look into it?”

What is missing?

Who? What does disengaged mean? Compared with what? What decision will follow?

THINK • PAIR • SHARE

Take 90 seconds: rewrite this concern as a question that data could genuinely help answer.

Four parts make a question analysable

Population

Who is the conclusion about?

2026 entry cohort

Variables

What will you observe?

weeks 1–4 submissions

Period

When does it apply?

first four teaching weeks

Purpose

What decision follows?

target support or change an intervention

TRY IT

What population, variable, period, and decision would you use to study “students are struggling with programming”?

WORKED EXAMPLE

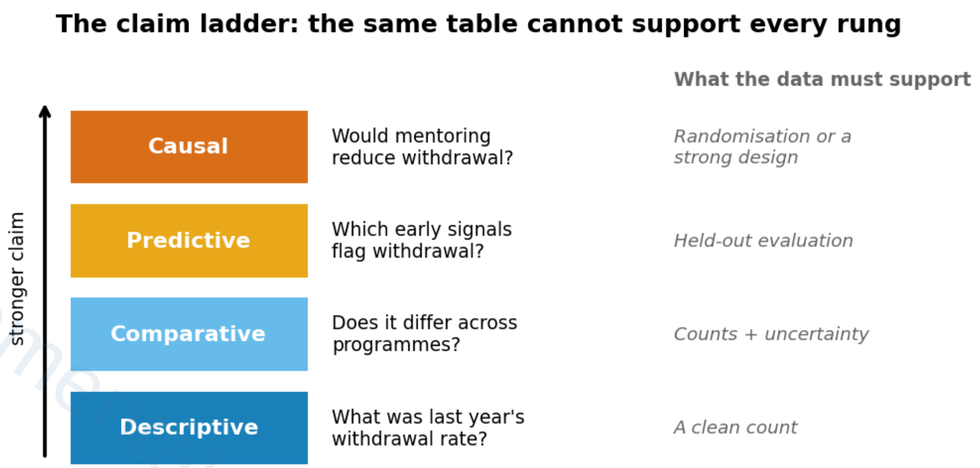
Repair the concern in four passes

1. Who?	Restrict to the 2026 entry cohort: 431 students.
2. What?	Define engagement explicitly: submissions in weeks 1–4.
3. Compared with what?	Compare with the 2025 cohort at the same point.
4. What decision?	If a difference concentrates, target support by programme.

Repaired question: Is mean assessment submission lower than in 2025, and does any difference concentrate in particular programmes?

Not every question needs—or permits—the same evidence

Descriptive	What was last year’s withdrawal rate?
Comparative	Does it differ across programmes?
Predictive	Which early indicators predict withdrawal?
Causal	Would mentoring reduce withdrawal?



IMPORTANT DISTINCTION

Which claim type is supported by a simple correlation between logins and grades?

CLAIM CHECK

Association is not an intervention result

Draft sentence

"Students who log in more often get better grades, so the university should require weekly logins."

Supported by observational data

Total logins and mean grade are positively associated ($r = 0.68$, $n = 1{,}200$).

Not supported

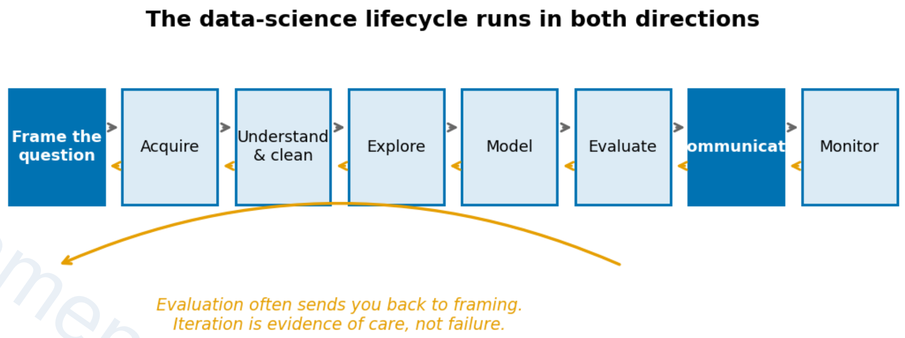
Requiring logins would raise grades. That is a causal claim and needs a different design.

VOTE

Name one alternative explanation for the association besides "logging in causes learning."

Good analysis loops; it does not move in a straight line

- Frame the question before touching the data
- Acquire and understand the data before cleaning
- Explore before modelling; evaluate before communicating
- Return to an earlier step when the evidence exposes a weak assumption



DISCUSS

Why might an evaluation result send you back to the framing or acquisition stage?

ONE-MINUTE PAUSE

What has to be true before a model is useful?

A fast model can still answer the wrong question. A correct calculation can still be irrelevant to a decision.

Write one sentence:

“Before I model, I need to know ________.”

SHARE

Ask the person beside you: did you name a data property, a decision, an ethical risk, or something else?

Rows identify units; columns describe attributes

A dataset is a representation of a process—not the process itself.

One row is one …

student
student-course registration
student-week of platform activity

Columns are variables

programme • study_mode • entry_score
logins • submissions •
minutes_on_platform

SAY IT ALOUD

Before any count, mean, or groupby: “One row of this table is one ______.”

MEET THE DATASET

One student, three different tables

Rows reflect different units of observation.

students.csv

student_id | programme | entry_score

100001 | CS | 72

100002 | Psychology | 65

enrolments.csv

student_id | course | grade

100001 | DS100 | 70

100001 | MATH110 | 65

activity.csv

student_id | week | logins

100001 | 1 | 8

100001 | 2 | 7

Student 100001 appears once in students, twice in enrolments, and once per week in activity.

CHOOSE THE TABLE

To count people, which table would you start from—and why?

Storage type is not the same as meaning

Nominal categories with no order

programme, study_mode

Ordinal ordered values; gaps may differ

midterm_satisfaction (1–5)

Discrete countable quantities

logins, submissions

Continuous any value in a range

entry_score, minutes

CHECK

student_id is stored as an integer. Why is its mean meaningless?

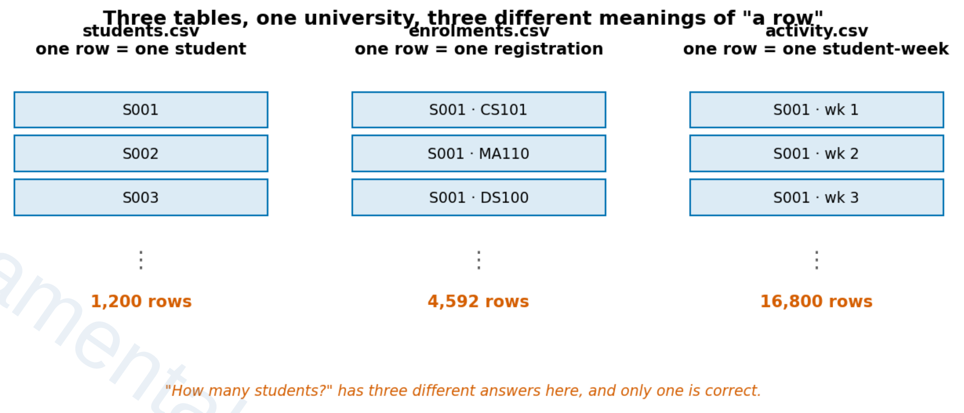
WORKED EXAMPLE

One row, one meaning

A colleague runs len(enrolments) and reports:

```
Number of students: 4665
```

But enrolments has one row per student-course registration—not one row per student.



PREDICT

How many distinct students are there? What line of code would you use to check?

THE ERROR IS SILENT

Correct code can answer the wrong question

4665

registration rows

1200

distinct students

3.9×

overcount

The remedy: if the noun in your question is “student,” aggregate to one row per student before you count.

TURN AND TALK

When might you deliberately want the registration-level table instead of the student-level table?

UNITS CHANGE RESULTS

The wrong unit can create systematic bias

Part-time share after joining enrolments

11.9%

one vote per registration

Part-time share in students table

14.4%

one vote per student

Students with fewer registrations are under-weighted in the first calculation.

EXPLAIN

Why is this not a random error that disappears when we collect more data?

SHORT BREAK

When we return: where data came from changes what it can mean

Before the break, write down one question you still have about:
claims • rows • variables • reproducibility

Return ready to challenge a number that looks convincing.

Every dataset carries the choices of the system that generated it

Files	CSV, JSON, Excel
APIs	services expose selected records
Databases	operational records from a process
Sensors & surveys	measurement and response choices
System exhaust	logs designed for operations—not your question

QUESTION

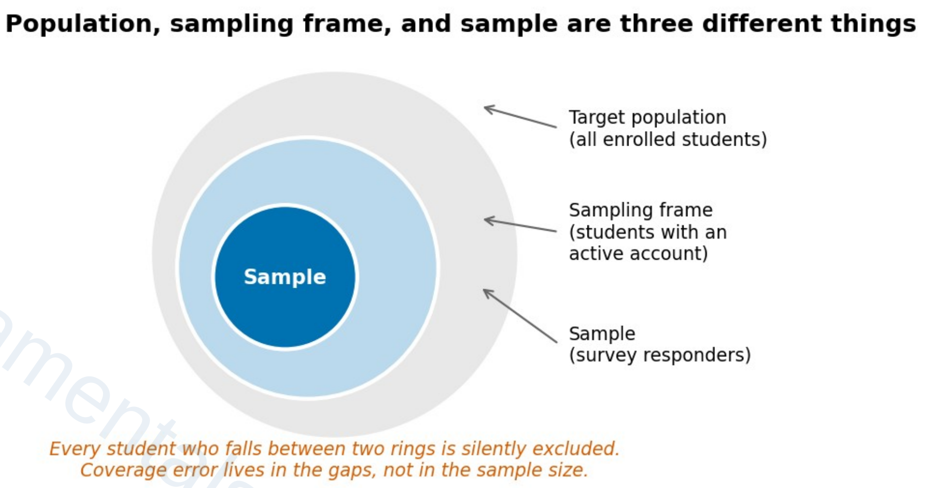
activity.csv records platform activity. Does that mean it measures engagement? What can it not see?

Population, sampling frame, and sample are different groups

Target population — everyone your conclusion is meant to describe

Sampling frame — the list you can actually draw from

Sample — the people who end up in your data



APPLY IT

For a platform satisfaction survey, who might be excluded before anyone chooses whether to respond?

A large convenience sample can be precisely wrong

Simple random	each unit equally likely	unbiased, needs a complete frame
Stratified	random within groups	protects small-group coverage
Cluster	sample whole groups	cheaper, but more uncertainty
Convenience	whoever is available	fast; bias of unknown direction

VOTE

Which is more credible: 200 randomly selected students or 5,000 volunteers? What assumption changes your answer?

TURN CONCEPTS INTO COLUMNS

An operational definition must be defended

Concept	Operational definition	Possible failure
Engagement	submissions in weeks 1–4	measures compliance, not learning
Academic difficulty	a stated rule using early evidence	may flag students who need a different kind of support
Satisfaction	survey response (1–5)	non-responders may differ systematically

SMALL GROUP TASK

Give two operational definitions of “engagement.” Which one would you trust more, and why?

Reliable is not the same as valid

Reliability

Would the measurement give the same value if measured again?

Login counts are usually reliable.

Validity

Does it measure the concept—or only something related to it?

Minutes on a platform measure a browser tab, not attention.

CHALLENGE

Name one student behaviour that produces low platform minutes but does not mean disengagement.

Ask who is mismeasured before operationalising a proxy

Part-time students show:

26.5% fewer

minutes on platform

26.8% fewer

logins

1.2% fewer

submissions

If all three predict withdrawal equally well, submissions are fairer and easier to explain than minutes or logins.

DEBATE

Should a model use the strongest proxy, the fairest proxy, or both? What evidence would you need?

MERGE WITH A HYPOTHESIS

State the expected cardinality before every join

students

one row per student

enrolments

one row per student-course

activity

one row per student-week

Before merging enrolments with students, expect many-to-one on student_id. The row count should stay the same.

PREDICT

If a merge changes 4,665 rows to 9,330 rows, what is the most likely data problem?

Let pandas test your merge assumption


```
before = len(enrolments)
merged = enrolments.merge(
    students, on="student_id", how="left",
    validate="many_to_one"
)
assert len(merged) == before
```

validate= raises an error if students contains duplicate student IDs.

The assertion catches unintended row growth—but you should also check for unmatched IDs.

READ THE CODE

What assumption does validate="many_to_one" make about each table?

A data dictionary and provenance record are part of the analysis

Data dictionary

name • type • meaning • units • permitted values • missingness • quirks

Provenance

source • collector • date • consent or licence • transformations

WHY IT MATTERS

Six weeks from now, what will you wish you had written down today?

A notebook is a transparent argument

01	**Markdown**	states the purpose, assumptions, and decisions
02	**Code**	performs transformations
03	**Output**	supplies evidence
04	**Interpretation**	connects evidence back to the question
05	**Restart and run all**	proves the analysis works from a clean state

SELF-CHECK

Could a classmate restart your notebook, understand every output, and reproduce the same conclusion?

RESPONSIBLE FRAMING

Ethics shapes the question before the model exists

Before collecting or using data about people, ask:

- Who benefits?
- Who bears the risk if we are wrong?
- Is participation voluntary?
- Could a less intrusive analysis meet the purpose?
- Is this variable for support, auditing, or a decision feature?
- Can the affected person understand and contest the outcome?

COMMITMENT

Complete this sentence: "A prediction of difficulty should be used to ________, not to ________."

EXIT TICKET

Show your Week 1 thinking

Before you leave, write four short answers:

1. Turn “students are struggling” into an analysable question.
2. Label one claim: descriptive, comparative, predictive, or causal.
3. Finish: “One row of enrolments.csv is one ______.”
4. Name one limitation or fairness risk in using platform logs as evidence.

NEXT STEP

Bring these answers to Lab 1—then test your claims directly in a reproducible notebook.

Your first transparent inspection

Before the lab

Set up Google Colab, place the seven CSV files in MyDrive/fds/data/, and run the course setup cell.

In the lab, you will

- Load the four tables and state what one row means in each
- Verify IDs and table structure in code
- Write one descriptive question, one predictive question, and one question this dataset cannot answer
- Restart and run all; submit a notebook plus a 150-word reflection

CARRY THIS RULE WITH YOU

Before any count, mean, groupby, or merge: state the unit of observation.